

consider the code shown below:

```
1. import cv2
2. img = np.random.randint(0, 256,
size=(512, 512, 3))
3. cv2.imwrite('img.jpg', img)
```

Line 1 imports the library OpenCV. Again, be careful, as the library name for the *import* statement is not *opencv*, unlike most other packages. Line 3 converts the matrix *img* to an image called *img.jpg*. Figure 10 shows this image generated by OpenCV. If you run this code in your system, the image will be generated in the same directory where your Jupyter Notebook is locally stored. If you check the properties of this image, you will see that it has a height of 512 pixels and width of 512 pixels. From these examples, it is easy to see that any AI or machine learning application that deals with computer vision uses a lot of arrays, vectors and matrices as well as ideas from linear algebra. Hence, our extensive treatment of arrays, vectors and matrices in this series is well justified.

Finally, consider the code shown below. What will the output image called *image.jpg* look like? I will give you two hints. The function *zeros* in lines 4 and 5 creates two 512 x 512 arrays *green* and *blue* filled with zeros. Lines 7 to 9 fill the three-dimensional array *img1* with values from the arrays *red*, *green*, and *blue*.

```
1. import numpy as np
2. import cv2
3. red = np.random.randint(0, 256,
size=(512, 512))
4. green = np.zeros([512, 512],
dtype=np.uint8)
5. blue = np.zeros([512, 512],
dtype=np.uint8)
6. img1 = np.zeros([512,512,3],
dtype=np.uint8)
7. img1[:, :,0] = blue
8. img1[:, :,1] = green
9. img1[:, :,2] = red
10. cv2.imwrite('image.jpg', img1)
```

```
print(np.random.randint(0, 256, size=(3, 3)))
print(np.random.randint(0, 256, size=(3, 3)))

[[170 212 84]
 [174 108 84]
 [236 196 178]]
[[152 168 106]
 [142 93 25]
 [ 45 20 78]]
```

Figure 9: Two matrices with random entries

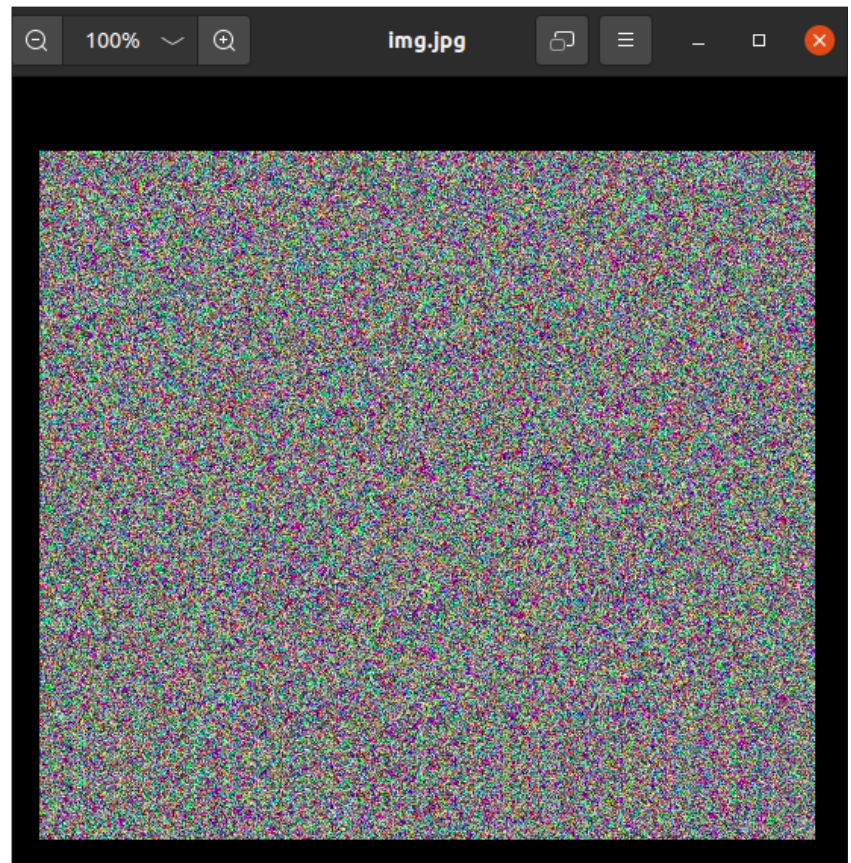


Figure 10: The image generated by OpenCV

We will wind up our discussion for now. In the next article in this series, we will begin by briefly learning about tensors, followed by installing and using a very powerful library called TensorFlow. TensorFlow is a serious player in the

world of AI and machine learning. Afterwards, we will take a break from matrices, vectors, and linear algebra for a short while and start learning a probability theory. If linear algebra is the brain of AI then probability is its heart, or vice versa. **END** 🐧

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